

Multi-source land-use emissions reveal rising airborne fraction

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Abstract The airborne fraction is the share of anthropogenic carbon dioxide emissions that remains in the atmosphere and is a key indicator of carbon-cycle response under continued emissions. Whether this share is rising remains debated because inference is sensitive to uncertainty in land-use and land-cover change (LULC) emissions. Here we use all available LULC measurement series from Global Carbon Budget 2025 data and estimate trends with a mixed-effects model with random intercepts and slopes by LULC series. We find that airborne fraction increased over 1959-2024, and that this conclusion is robust to excluding the final year and to a measurement-error-aware weighted least squares robustness specification. These results strengthen evidence that an increasing share of emitted carbon dioxide is accumulating in the atmosphere rather than being taken up by land and ocean sinks, with implications for carbon-budget assessment and near-term mitigation requirements.

1 Background

2 Whether the airborne fraction (AF) is increasing or approximately constant remains contested¹⁻⁷. In
3 its classical form, AF is a yearly ratio of atmospheric growth to total anthropogenic emissions,
4 computed as the sum of fossil fuel emissions and land-use and land-cover change emissions as

$$AF_t = \frac{G_t}{FF_t + LULC_t}, \quad (1)$$

5 where G_t is the annual atmospheric CO_2 growth, FF_t is fossil fuel emissions excluding carbonation,
6 and $LULC_t$ is land-use and land-cover change emissions. AF is a key carbon-cycle diagnostic, with
7 implications for carbon-cycle feedbacks and near-term mitigation planning^{1,2,8}.

8 A persistent concern is that AF inference depends on the treatment of land-use and land-cover
9 change (LULC) emissions, which are uncertain and model-dependent in annual carbon-budget
10 accounting. The Global Carbon Budget (GCB) 2025⁸ provides one column of LULC emissions as the

11 average of three bookkeeping models (BLUE⁹, OSCAR¹⁰, LUCE¹¹), but a broader set of model-based
12 LULC alternatives can be constructed from the same source. Estimates that do not incorporate this
13 multi-source LULC information can be underpowered and inference on trend direction becomes less
14 reliable.

15 Here we address that issue with a statistical design that uses all repeated LULC measurements in a
16 mixed-effects trend framework to obtain more reliable inference. The main specification estimates AF
17 trends from a panel of yearly AF values constructed from each LULC measurement series, with
18 random intercepts and random slopes capturing between-series heterogeneity and inducing within-
19 series dependence over time. As a robustness exercise, we also estimate a two-stage measurement-
20 error weighted least squares (WLS) model that propagates denominator uncertainty from repeated
21 LULC measurements into annual AF uncertainty and uses heteroskedasticity-and-autocorrelation-
22 consistent (HAC) inference^{12–14}.

23 Across specifications that incorporate the multi-source LULC information, we find robust evidence
24 that AF has increased over 1959-2024, and that this conclusion is not driven by the final observation
25 (2024), which shows a large AF value. By contrast, ordinary least squares (OLS) specifications that do
26 not incorporate denominator measurement information provide weaker evidence of a positive trend
27 and are more sensitive to endpoint exclusion. Taken together, the results clarify why AF trend
28 significance has been elusive in the literature and strengthen evidence that a growing share of
29 emitted carbon dioxide is accumulating in the atmosphere.

30 **Data**

31 We use annual Global Carbon Budget 2025 data for 1959-2024, with atmospheric growth G_t from
32 NOAA/ESRL global concentration trends¹⁵, fossil emissions excluding carbonation FF_t from the
33 Global Carbon Project fossil dataset⁸, and a panel of 69 LULC measurements per year: BLUE⁹,
34 OSCAR¹⁰, LUCE¹¹, and peat-augmented^{16–18} process-based land-model combinations drawn from the
35 GCB model ensemble^{19–39} (see Methods). The data is shown in Figure 1.

36 Two LULC means can be constructed from the panel. First, the GCB LULC mean, defined as the mean
37 of the three bookkeeping models (BLUE, OSCAR, and LUCE), corresponds to the LULC column used
38 in the Global Carbon Budget. Second, the all-LULC mean, defined as the cross-series mean across all
39 69 LULC measurements. The literature has typically used the GCB LULC mean as the denominator in
40 AF trend estimation, but we show that incorporating the full set of LULC measurements increases the
41 precision of trend estimates and provides more robust inference on trend direction.

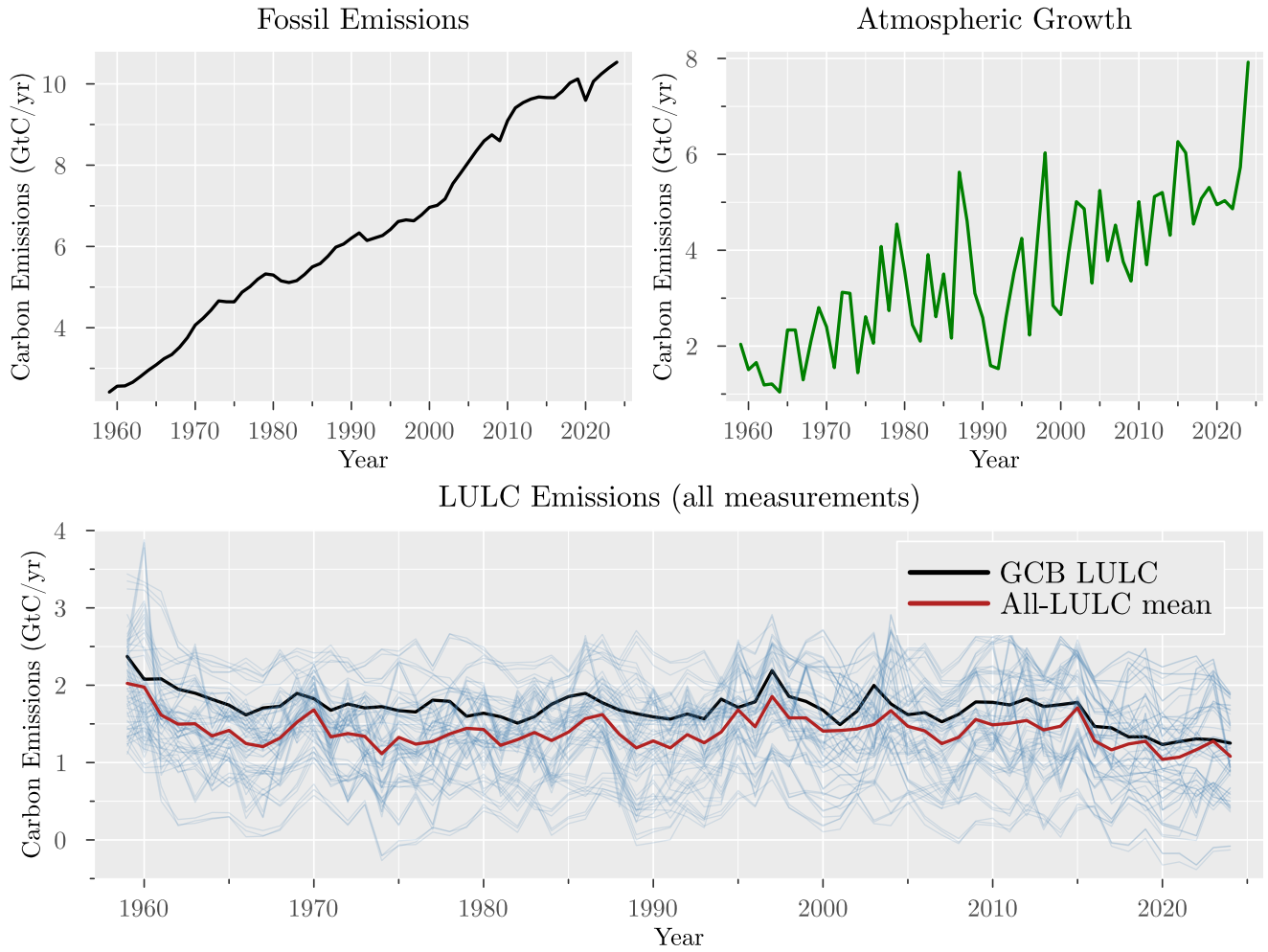


Figure 1: Global Carbon Budget 2025 data for fossil emissions, atmospheric growth, and LULC emissions (1959-2024). For the LULC panel, we show all extracted and derived LULC measurements together with the GCB LULC column and the all-LULC cross-series mean.

Identification strategy

The empirical question is whether AF has a positive linear trend. The key design choice is how to use the full panel of repeated LULC measurements while accounting for within-series dependence and cross-series heterogeneity. OLS on a collapsed annual series discards panel information and hence provides less reliable inference, see results in Table 2. Our preferred specification is a mixed-effects model with random intercepts and random slopes by LULC series, allowing correlation across years within each series and capturing unobserved heterogeneity in LULC measurements^{40,41} (see Methods). For each LULC measurement series, we construct a series of AF estimates by year (Equation 1), and the model estimates a common time trend across all series while allowing for series-specific intercepts and slopes. Formally, the model can be written as

$$AF_{t,j} = \alpha_j + \beta_j t + \varepsilon_{t,j},$$

56 where

$$\alpha_j \sim N(\mu_\alpha, \sigma_\alpha^2), \quad \beta_j \sim N(\mu_\beta, \sigma_\beta^2), \quad \varepsilon_{t,j} \sim N(0, \sigma_\varepsilon^2).$$

57 Here μ_α and μ_β are the fixed effects (overall intercept and slope), while σ_α^2 and σ_β^2 capture the
58 variability in intercepts and slopes across LULC measurement series. The error term $\varepsilon_{t,j}$ captures
59 idiosyncratic noise. The key parameter of interest is μ_β , which captures the average trend across all
60 LULC measurement series. Incorporating the full panel of LULC measurements in this way allows us
61 to obtain more precise and robust estimates of the AF trend, while accounting for measurement
62 uncertainty and heterogeneity across LULC series.

63 Results

64 Main estimates

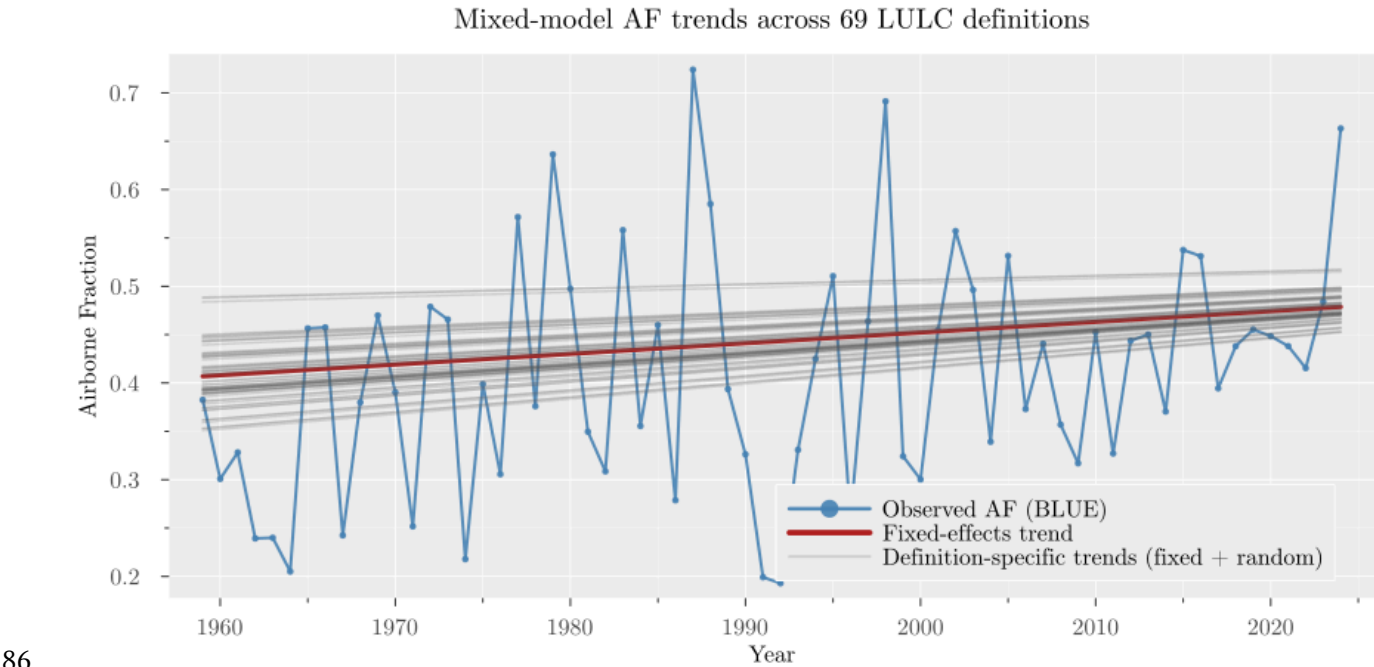
65 The main results from estimating the mixed-effects model over the full sample (1959-2024) are shown
66 in Table 1, and the associated trend lines are shown in Figure 2a.

67 Table 1: Mixed-effects model with random effects for each LULC measurement series.

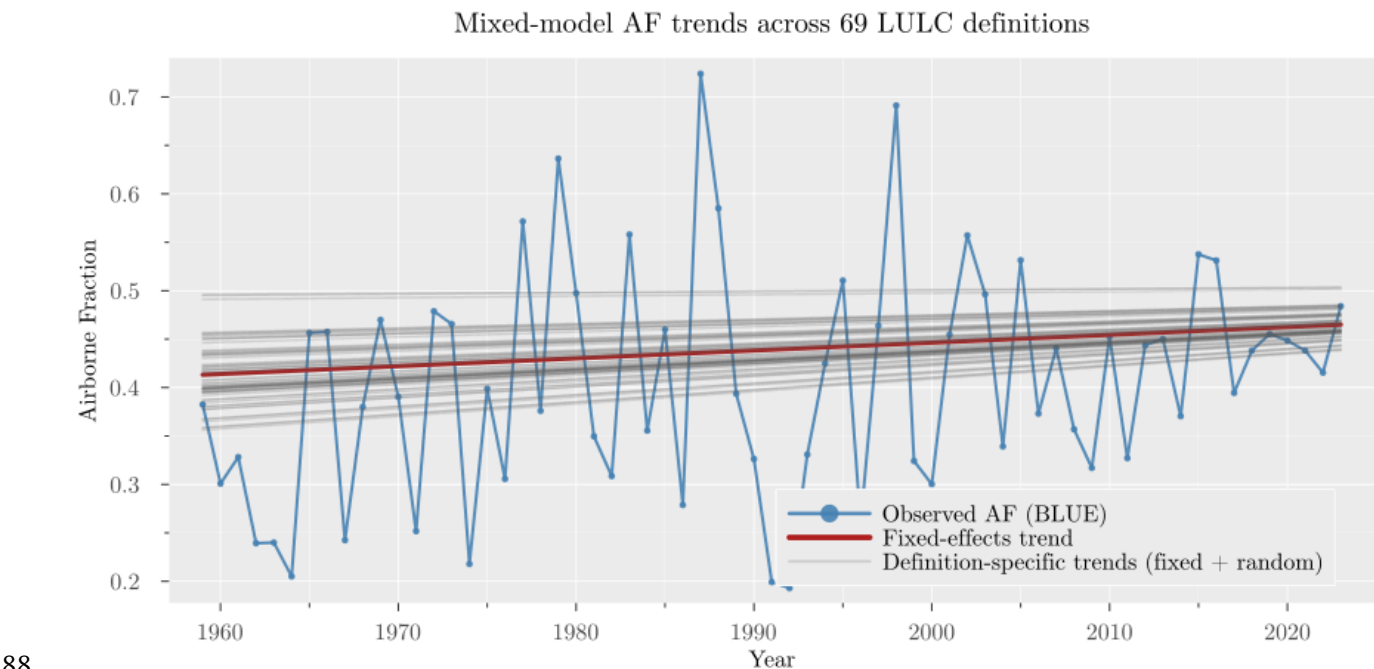
68 Parameter	Intercept (full)	Slope (full)	Intercept (up to 69 2023)	Slope (up to 2023)
70 Estimate	0.406959	0.001103	0.413260	0.000808
71 Standard	0.005573	0.000102	0.005586	0.000103
72 error				
73 p-value	1.0E-5	1.0E-5	1.0E-5	1.0E-5
74 R-squared	0.008631	0.008631	0.004518	0.004518
75 (marginal)				
76 R-squared	0.700967	0.700967	0.708173	0.708173
77 (conditional)				

78 The mixed-effects slope is positive and statistically significant (p-value < 1.0E-5), indicating strong
79 evidence of an increasing AF trend across LULC measurement series. The estimated intercept is
80 around 0.407+/-0.006 and the slope is around 0.0011+/-0.0001 per year, implying an increase of about
81 0.0717 in AF over 1959-2024. These estimates are comparable with the approximately 0.45 AF level
82 reported in the literature^{2,3,6,7,42,43}, while identifying a positive long-run trend when all LULC
83 information is incorporated. The conditional R-squared values indicate that most explained variation

84 is captured by between-series structure rather than by the fixed linear time trend (marginal versus
85 conditional R-squared).



(a) AF series with mixed-effects model trend (full sample, 1959-2024).



(b) AF series with mixed-effects model trend (sample ending in 2023).

Figure 2:

91 Figure 2a shows the AF series for each LULC measurement together with the overall mixed-effects
 92 trend line. The figure shows a clear positive trend across the panel of AF series, with the mixed-
 93 effects model capturing this common signal while allowing for series-specific heterogeneity.

94 **Endpoint and model specification robustness**

95 AF shows a large value in the final year (2024), so we re-estimate the mixed-effects model on the
 96 subsample ending in 2023 to check whether the result is driven by that observation. Results are
 97 shown in the last two columns of Table 1 and Figure 2b. The slope remains positive and statistically
 98 significant (p-value < 1.0E-5), indicating that the increasing AF trend is not driven by the final
 99 observation. The slope is smaller (0.000808 per year versus 0.001103 per year), consistent with a
 100 contribution from the 2024 large value, but the direction and inference are unchanged.

101 As an additional robustness check, we estimate a delta-method uncertainty proxy for each annual AF
 102 estimate using the cross-measurement dispersion of the full 69-series LULC panel, and use these
 103 proxies in WLS with HAC inference^{12–14,44}. Two WLS specifications are shown in Table 2: one using
 104 the GCB LULC mean as the denominator and one using the all-LULC mean. In both specifications,
 105 weights are constructed from the all-LULC panel dispersion so that the weighting scheme is held
 106 fixed across denominator definitions.

107 The WLS results are consistent with the mixed-effects results and show a positive, statistically
 108 significant trend. Note that the slopes estimated from WLS are steeper than those from OLS or the
 109 mixed-effects model, which is consistent with the fact that WLS assigns less weight to years with
 110 higher denominator uncertainty, which tend to be the earlier years of the sample.

Table 2: Trend comparison across OLS and WLS specifications for the full sample and the sample ending in 2023.

Trend	OLS (GCB)	OLS (all) full	WLS (GCB)	WLS (all) full	OLS (GCB)	OLS (all) up to 2023	WLS (GCB)	WLS (all) up to 2023
Estimate	0.00158	0.00117	0.00265	0.00230	0.00130	0.00088	0.00237	0.00202
Standard error	0.00077	0.00079	2.0E-5	2.0E-5	0.00078	0.00080	2.0E-5	2.0E-5
HAC standard error	0.00062	0.00064	0.00065	0.00066	0.00058	0.0006	0.00059	0.0006
p-value	0.04021	0.13891	1.0E-5	1.0E-5	0.09516	0.27109	1.0E-5	1.0E-5
HAC p-value	0.01112	0.06846	4.0E-5	0.00050	0.02495	0.14322	5.0E-5	0.00080
R-squared	0.06171	0.03309	0.14899	0.11107	0.04233	0.01886	0.12552	0.08930

By contrast, OLS specifications that do not incorporate denominator measurement information show weaker evidence of a positive trend and greater endpoint sensitivity. In the full sample, OLS using GCB LULC is significant under both conventional and HAC inference (p-value 0.04021 and HAC p-value 0.01112), whereas OLS using the all-LULC mean is not significant under either method (p-value 0.13891 and HAC p-value 0.06846).

In the sample ending in 2023, WLS remains significant under both mean definitions (HAC p-values 5.0E-5 and 0.00080), whereas OLS only remains significant when using the GCB LULC mean (HAC p-values 0.02495 and 0.14322).

This sensitivity to denominator construction and endpoint exclusion highlights the value of incorporating multi-source LULC information in AF trend estimation.

Discussion and conclusion

Using a framework that incorporates all available LULC information from Global Carbon Budget 2025, we find robust evidence that AF increased from 1959 to 2024. Methodologically, incorporating

141 multi-source uncertainty materially changes inference relative to plain OLS. Endpoint tests show that
 142 this conclusion is not driven by the large value in 2024.
 143 A rising AF implies that a larger share of emitted CO_2 remains in the atmosphere over policy-
 144 relevant horizons, tightening near-term mitigation requirements for a given temperature
 145 objective^{1,2,8}. The increasing AF signal is consistent with broader evidence that the climate system is
 146 out of energy balance and has recently shown elevated warming and heating rates^{45–48}; these findings
 147 should be interpreted in that wider risk context.

148 **Methods**

149 Our primary estimator is a linear mixed-effects trend model with random intercepts and random
 150 slopes for each LULC measurement series^{40,41,49,50}. As a robustness check, we also estimate a
 151 measurement-error-aware airborne fraction variance using repeated yearly denominator
 152 measurements, followed by WLS trend estimation with HAC inference^{12–14,44,51}.

153 **Construction of the LULC measurement panel**

154 The repeated LULC measurements are built from the Global Carbon Budget 2025 dataset. We first
 155 extract the three bookkeeping series: BLUE⁹, OSCAR¹⁰, LUCE¹¹. Then, for each process-based land-
 156 model LULC series in the GCB ensemble that does not already include peat emissions^{19–39}, we add the
 157 corresponding peat component to make it comparable to the bookkeeping models, which include
 158 peat emissions by construction. Specifically, for each of the 22 process-based series we create three
 159 peat-augmented variants by adding FAO_peat, LPX_Bern_peat, and ORCHIDEE_peat^{16–18}. This
 160 produces 66 derived series, and together with BLUE/OSCAR/LUCE gives a panel of 69 yearly LULC
 161 measurements.

162 **Mixed-effects model**

163 The primary estimator is a linear mixed-effects trend model fitted on the panel of yearly AF values
 164 constructed from each LULC measurement series. For each year t and series j , we define

$$AF_{t,j} = \frac{G_t}{FF_t + LULC_{t,j}},$$

165 and estimate

$$AF_{t,j} = (\mu_\alpha + u_{\alpha j}) + (\mu_\beta + u_{\beta j})t + \varepsilon_{t,j},$$

166 with

$$\begin{bmatrix} u_{\alpha j} \\ u_{\beta j} \end{bmatrix} \sim \left(N \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma_u \right), \quad \varepsilon_{t,j} \sim N(0, \sigma_\varepsilon^2).$$

167 This specification allows each LULC series to have its own level and trend while estimating a
 168 common population-average trend μ_β . The main estimand is μ_β , interpreted as the average annual
 169 change in AF across the full set of LULC definitions. Note that this notation is equivalent to the more
 170 compact notation in the main text, where $\alpha_j = \mu_\alpha + u_{\alpha j}$ and $\beta_j = \mu_\beta + u_{\beta j}$.
 171 The model is estimated by restricted maximum likelihood^{40,41}. Fixed-effect uncertainty is based on
 172 model-based standard errors and Wald tests. We report the estimated fixed intercept and slope,
 173 standard errors, p-values, and marginal and conditional R-squared^{49,50}.
 174 Identification and interpretation rely on standard mixed-model conditions: (i) conditional linearity in
 175 time, (ii) random effects centred at zero and independent across LULC definitions, and (iii) mean-zero
 176 residuals conditional on fixed and random effects.
 177 As an endpoint robustness check, we re-estimate the same mixed-effects specification on the sample
 178 ending in 2023 to verify that inference on μ_β is not mechanically driven by the final-year
 179 observation.

180 **Weighted least squares estimation approach**

181 As a robustness check to the mixed-effects model, we also estimate a two-stage measurement-error-
 182 aware WLS model that uses repeated LULC measurements to construct a denominator-uncertainty
 183 proxy for each year and HAC inference^{12–14,44,51}.
 184 In the estimation, the yearly denominator is computed as

$$\hat{C}_t = FF_t + \overline{LULC}_t, \quad \overline{LULC}_t = \frac{1}{n_t} \sum_{j=1}^{n_t} LULC_{tj},$$

185 where $n_t = 3$ for the GCB LULC mean and $n_t = 69$ for the all-LULC mean. In the implemented
 186 robustness specification, the yearly weighting term is computed from the cross-measurement
 187 dispersion across the full set of 69 LULC values and is then applied to both denominator definitions.
 188 Assume for each time $t = 1, \dots, T$ we observe:
 189 • a single numerator G_t (e.g., atmospheric CO_2 growth), and
 190 • multiple denominator measurements $c_{t1}, \dots, c_{tn_t}$ with $n_t \geq 2$, which are noisy observations of a
 191 latent C_t .

Using the repeated denominator measurements, we construct a year-specific uncertainty proxy for the ratio estimator $AF_t = G_t/C_t$ via the delta method, which captures how dispersion in denominator definitions propagates into AF. The procedure is described in detail next.

Step 1: Estimate C_t and AF_t

1. Construct the denominator estimate at time t as the sample mean across available LULC measurements as

$$\hat{C}_t = \frac{1}{n_t} \sum_{j=1}^{n_t} c_{tj}.$$

We consider two variants of this denominator estimate: one using the GCB LULC column (the mean of BLUE, OSCAR, LUCE) and one using the all-LULC mean (the mean across all 69 LULC measurements).

The cross-measurement dispersion around $\hat{C}_t$ is estimated from the multiple measurements and used as a weighting proxy.

With $n_t \geq 2$, the empirical dispersion proxy is

$$\widehat{\text{Var}}(\hat{C}_t) = s_{c,t}^2, \quad s_{c,t}^2 = \frac{1}{n_t - 1} \sum_{j=1}^{n_t} (c_{tj} - \bar{c}_t)^2, \quad \bar{c}_t = \hat{C}_t.$$

2. Form the ratio estimate

$$AF_t = G_t/\hat{C}_t.$$

3. Use the multivariate delta method (shown below) to obtain an estimate of the AF_t uncertainty

$$\widehat{\text{Var}}(AF_t) = \left(\frac{G_t}{\hat{C}_t^2} \right)^2 s_{c,t}^2.$$

Step 2: WLS regression of AF_t on time

For each time t , we obtain a point estimate AF_t and an uncertainty proxy that reflects cross-measurement dispersion. We use these quantities to fit a linear trend via WLS:

$$AF_t = \alpha + \beta t + \varepsilon_t, \quad \varepsilon_t \sim (0, \sigma_t^2), \quad \sigma_t^2 = \text{Var}(AF_t).$$

In our application, the time index t is the year, and AF_t is the estimated AF for that year. The weighting term is obtained from AF uncertainty from denominator dispersion and varies across years with the spread of the LULC measurement panel. We use this term to weight the regression, giving more weight to years with lower denominator-related uncertainty. The core idea is that years with

213 more consistent LULC measurements provide more reliable AF estimates, and hence should have
 214 greater influence on the trend estimation.

215 Assuming negligible correlation in ε_t across time, this is WLS with weights $w_t = \frac{1}{\sigma_t^2}$. If we suspect
 216 serial correlation, we can use HAC standard errors for inference on β without changing the point
 217 estimate. In practice, we use a HAC covariance estimator with a Bartlett kernel and Andrews
 218 automatic bandwidth selection¹⁴. In the results, we report both conventional and HAC standard
 219 errors.

220 Testing for a time trend is then a test of $\beta = 0$ versus $\beta \neq 0$ in this linear model, and WLS uses the
 221 repeated denominator information through this delta-method weighting scheme.

222 **Delta method for ratio variance estimation**

223 **Step-by-step derivation (first-order delta method):**

224 Define the random vector and the function of interest as

$$X_t = \begin{bmatrix} G_t \\ \hat{C}_t \end{bmatrix}, \quad g(X_t) = \frac{G_t}{\hat{C}_t}.$$

225 and let

$$\sigma_{G,t}^2 = \text{Var} (G_t), \quad \sigma_{\hat{C},t}^2 = \text{Var} (\hat{C}_t), \quad \sigma_{GC,t} = \text{Cov} (G_t, \hat{C}_t).$$

226 1. Linearize g around the mean vector $(\mu_{G,t}, \mu_{C,t}) = \mathbb{E}[(G_t, \hat{C}_t)]$:

$$g(X_t) \approx g(\mu_{G,t}, \mu_{C,t}) + \nabla g(\mu_{G,t}, \mu_{C,t})^\top (X_t - \mathbb{E}[X_t]).$$

227 2. Compute the gradient:

$$\nabla g(G, C) = \begin{bmatrix} \partial g / \partial G \\ \partial g / \partial C \end{bmatrix} = \begin{bmatrix} 1/C \\ -G/C^2 \end{bmatrix}.$$

228 3. Write the covariance matrix of $(G_t, \hat{C}_t)$:

$$\Sigma_t = \begin{bmatrix} \sigma_{G,t}^2 & \sigma_{GC,t} \\ \sigma_{GC,t} & \sigma_{\hat{C},t}^2 \end{bmatrix}.$$

229 4. Apply the delta-method variance formula

$$\text{Var} (g(X_t)) \approx \nabla g(\mu_{G,t}, \mu_{C,t})^\top \Sigma_t \nabla g(\mu_{G,t}, \mu_{C,t}),$$

230 and with plug-in evaluation at (G_t, C_t) , this becomes

$$\text{Var} (AF_t) \approx \left(\frac{1}{C_t} \right)^2 \sigma_{G,t}^2 + \left(\frac{G_t}{C_t^2} \right)^2 \sigma_{\hat{C},t}^2 - 2 \frac{G_t}{C_t^3} \sigma_{GC,t}.$$

231 In practice, we use plug-in estimates (replace unknown moments by their empirical counterparts).
 232 This is the standard delta-method approximation for the variance of a ratio estimator.
 233 5. In the case analysed in this paper, we have a single numerator measurement from a robust source
 234 per time, so we treat $\sigma_{G,t}^2$ and $\sigma_{GC,t}$ as negligible relative to the variance from the denominator
 235 measurement error, which is captured by $\sigma_{C,t}^2$. Nevertheless, the formula above is general and can
 236 be applied in other contexts where numerator uncertainty is non-negligible or where there is
 237 covariance between numerator and denominator measurements.
 238 Under the assumption of negligible numerator variance and covariance, the formula simplifies to

$$\text{Var} (AF_t) \approx \left(\frac{G_t}{C_t^2} \right)^2 \sigma_{C,t}^2.$$

239 Accordingly, a practical plug-in variance estimator for AF_t is

$$\widehat{\text{Var}}(AF_t) = \left(\frac{G_t}{C_t^2} \right)^2 s_{c,t}^2.$$

240 **Data availability**

241 The study uses publicly available Global Carbon Budget 2025 data and derived yearly series
 242 generated from those source files. The processed analysis tables are provided in the manuscript
 243 repository under the results directory, and the extracted and derived LULC panel is available as a
 244 CSV file in the data directory.

245 **Code availability**

246 All code used to process data, estimate models, and generate figures is available in the repository
 247 (github.com/everval/Airborne-Fraction-WLS-Trend) under the scripts directory, including Quarto
 248 analysis files and Julia helper functions.

249 **Competing interests**

250 The author declares no competing interests.

251 **Additional information**

252 Supplementary information is available in the accompanying supplementary manuscript file.

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